JUNSU KIM

Curriculum Vitae

sam9732@kaist.ac.kr | Homepage | LinkedIn | GitHub

Research Interests

- Egocentric and embodied AI for tool use, physical interactions, and action-driven state changes.
- 3D/4D scene understanding and action-centric world models trained on human and robot data.

Education

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, South Korea
B.S. in Electrical Engineering Mar 2021 – *Expected Aug 2027*
GPA: 4.09/4.3 (Major GPA: 4.19/4.3)
Military Service, Republic of Korea Army: Jul 2022 – Jan 2024

Nanyang Technological University (NTU) Singapore
Exchange Student Jan 2026 – May 2026
School of Electrical and Electronic Engineering

Publications

[1] **EgoTools: Tool-Centric Reasoning in Real-World Egocentric Videos**.

Shulin Tian*, **Junsu Kim***, Shuai Liu, Hao Li, Yujiao Shen, Sihan Li, Zhe Yang, Yeongon Kim, Runmao Yao, Yuhao Dong, Fangzhou Hong, Antonino Furnari, Jingkang Yang, Hongyuan Zhu, Ziwei Liu.

EMNLP 2026 *(Main Conference)*.

Contribution: Co-led dataset and benchmark development, designed annotation and QA workflows, constructed the 1,000-question benchmark, fine-tuned Qwen3-VL-8B, and conducted baseline and generalization analyses.

[2] **C3G: Learning Compact 3D Representations with 2K Gaussians**.
[Project Page] [Paper].

Honggyu An*, Jaewoo Jung*, ..., Hyuna Ko, **Junsu Kim**, ..., Seungryong Kim.

CVPR 2026.

Contribution: Implemented analysis tools and conducted experiments for feed-forward 3D reconstruction pipelines, focusing on novel-view synthesis quality and compact 3D representation behavior.

*Equal contribution.

Research Experience

S-Lab (MMLab), NTU Jan 2026 – Present
Research Intern (Advisor: Prof. Ziwei Liu, NTU)

- Co-led **EgoTools**, a real-world egocentric dataset and benchmark for tool-centric reasoning, built from **100+ hours** of long-horizon tool-use videos.
- Designed annotation and QA workflows for physical-state, procedural, affordance, and tool-use reasoning; coordinated data collection and quality control; and co-led construction of a **1K-QA** benchmark.

- Fine-tuned **Qwen3-VL-8B** on **220K** multimodal training examples, improving EgoTools-Bench accuracy from **50.0%** to **63.9%**; conducted cross-benchmark generalization, failure-case, and catastrophic-forgetting analyses.

Computer Vision Lab (CVLab), KAIST AI Jun 2025 – Present
Research Intern (Advisor: Prof. Seungryong Kim, KAIST AI)

- *Jun 2026 – Present:* Investigating latent action representations for video dynamics modeling and action-conditioned future prediction.
- *Jun 2025 – Dec 2025:* Conducted feed-forward 3D reconstruction and novel-view synthesis research using Gaussian Splatting, VGGT, and NoPoSplat, focusing on geometry quality, failure cases, and compact 3D representation behavior.

Video and Image Computing Lab (VICLab), KAIST Dec 2024 – Jun 2025
Research Intern (Advisor: Prof. Mun Churl Kim, KAIST EE)

- Implemented and evaluated object-detection pipelines spanning two-stage, YOLO-style, and Vision Transformer architectures for few-shot and anomaly detection tasks.

Teaching Experience

MAS.10001 Calculus I Tutor, KAIST Freshman Tutoring Program Mar 2022 – Jun 2022
Supported freshman students through problem-solving sessions and conceptual review.

Honors & Scholarships

KAIST **Presidential Fellowship** (15th) Feb 2025 – Present
National Science & Engineering Scholarship for Academic Excellence Aug 2025 – Present
Woon Hae Scholarship (12th) Feb 2025 – Dec 2025
KAIST **Dean's List**, College of Engineering Fall 2021, Spring 2024
awarded to the top 3% of students
KAIST Shared Lives Award (Top-ranked Recipient) Feb 2026
For exceptional and sustained blood donation service

Leadership & Service

President, EERun (KAIST EE Running Club) Sep 2024 – Dec 2025
Organized weekly training sessions and semester events for KAIST EE students.
Google Student Ambassador Sep 2025 – Dec 2025
Promoted Gemini Pro to KAIST students through short-form video content.
Additional Activities 2022 – Present
KAIST AI Studying Club, Young Engineers Honor Society (YEHS), SilverLining, Freshman Proctor.

Skills

Programming	C/C++, Python, PyTorch, Git, LaTeX
Research Tools	video preprocessing, dataset curation, annotation pipelines, benchmark construction, evaluation scripts
Languages	Korean (native), English (fluent, iBT TOEFL 110)